# Derivation of Recurring Deposit Yield Formula

Mathematical Typesetting of Handwritten Notes (IMG_20261004_090254)

**Notation & Variables**

| | | |
|---|---|---|
| $I$ (or $A$) | : | Installment amount deposited at the end of every payment period $R_d$. |
| $R_d$ | : | Duration of each recurring deposit installment period (in days). |
| $C_d$ | : | Duration of each compounding period (in days). |
| $T_d$ | : | Total tenure / duration of the recurring deposit (in days). |
| $r$ | : | Annual rate of interest (%). |
| $Y_d$ | : | Total duration of one year (in days, e.g., 365). |

## Step-by-Step Mathematical Derivation

Below is the complete, line-by-line LaTeX transcription and typesetting of the mathematical derivation from IMG_20261004_090254.

### 1. Initial Compound Interest Summation

The total accrued value across all $K$ installment periods is formulated as the sum of individually compounded installments, adjusted by the annualized day ratio $\frac{Y_d}{T_d}$:

$$\left[\sum_{K=0}^{\left(\frac{T_d}{R_d}-1\right)} I\left(1+\frac{C_d}{Y_d}\frac{r}{100}\right)^{\frac{T_d-KR_d}{C_d}}\right]\times\frac{Y_d}{T_d} \tag{1}$$

### 2. Factoring Constant Terms

Decomposing the exponent $\frac{T_d-KR_d}{C_d}=\frac{T_d}{C_d}-\frac{KR_d}{C_d}$ allows splitting the compound term into product factors:

$$=\frac{Y_d}{T_d}\times I\sum_{K=0}^{\left(\frac{T_d}{R_d}-1\right)}\left(1+\frac{C_d}{Y_d}\frac{r}{100}\right)^{\frac{T_d}{C_d}}\left(1+\frac{C_d}{Y_d}\frac{r}{100}\right)^{-\frac{KR_d}{C_d}} \tag{2}$$

Factoring out the constant factor $\left(1+\frac{C_d}{Y_d}\frac{r}{100}\right)^{\frac{T_d}{C_d}}$ outside the summation yields:

$$=\frac{Y_d}{T_d}\times I\times\left(1+\frac{C_d}{Y_d}\frac{r}{100}\right)^{\frac{T_d}{C_d}}\sum_{K=0}^{\left(\frac{T_d}{R_d}-1\right)}\left(1+\frac{C_d}{Y_d}\frac{r}{100}\right)^{-\frac{KR_d}{C_d}} \tag{3}$$

### 3. Evaluating the Finite Geometric Series

The remaining summation represents a finite geometric series $\sum_{K=0}^{N-1}x^K=\frac{x^N-1}{x-1}$, where $x=\left(1+\frac{C_d}{Y_d}\frac{r}{100}\right)^{-\frac{R_d}{C_d}}$ and $N=\frac{T_d}{R_d}$:

$$= \frac{Y_d}{T_d} \times I \times \left(1 + \frac{C_d}{Y_d}\frac{r}{100}\right)^{\frac{T_d}{C_d}} \left[ \frac{\left[\left(1 + \frac{C_d}{Y_d}\frac{r}{100}\right)^{-\frac{R_d}{C_d}}\right]^{\frac{T_d}{R_d}} - 1}{\left(1 + \frac{C_d}{Y_d}\frac{r}{100}\right)^{-\frac{R_d}{C_d}} - 1} \right] \tag{4}$$

Simplifying the exponent in the numerator $\left(-\frac{R_d}{C_d}\right) \times \left(\frac{T_d}{R_d}\right) = -\frac{T_d}{C_d}$:

$$= \frac{Y_d}{T_d} \times I \times \left(1 + \frac{C_d}{Y_d}\frac{r}{100}\right)^{\frac{T_d}{C_d}} \left[ \frac{\left(1 + \frac{C_d}{Y_d}\frac{r}{100}\right)^{-\frac{T_d}{C_d}} - 1}{\left(1 + \frac{C_d}{Y_d}\frac{r}{100}\right)^{-\frac{R_d}{C_d}} - 1} \right] \tag{5}$$

## 4. Algebraic Simplification to Positive Exponent Form

Multiplying the numerator and denominator by positive exponential scaling factors eliminates negative exponents:

$$= \frac{Y_d}{T_d} \times I \times \left[ \frac{\left[1 - \left(1 + \frac{C_d}{Y_d}\frac{r}{100}\right)^{\frac{T_d}{C_d}}\right]\left(1 + \frac{C_d}{Y_d}\frac{r}{100}\right)^{\frac{R_d}{C_d}}}{\left[1 - \left(1 + \frac{C_d}{Y_d}\frac{r}{100}\right)^{\frac{R_d}{C_d}}\right]} \right] \tag{6}$$

**Final Typeset Closed-Form Result**

$$\text{Value} = \frac{Y_d}{T_d} \cdot I \cdot \frac{\left[1 - \left(1 + \frac{C_d \cdot r}{100 \cdot Y_d}\right)^{\frac{T_d}{C_d}}\right]\left(1 + \frac{C_d \cdot r}{100 \cdot Y_d}\right)^{\frac{R_d}{C_d}}}{1 - \left(1 + \frac{C_d \cdot r}{100 \cdot Y_d}\right)^{\frac{R_d}{C_d}}}$$